HYDRA

Turn website tasks into reusable JSON APIs

The idea

Hydra is an AI-powered platform that converts a natural-language browser task into a reusable API. A user provides a website URL, describes the information they need, and confirms they are authorized to access it. Hydra navigates the website, extracts the requested data, converts it into clean JSON, and publishes an endpoint that other applications can call.

Why it matters

Many useful websites have no public API. Teams must browse them manually or maintain a separate scraper for every source. Those scripts often fail on JavaScript-heavy pages, complex navigation, or layout changes. Hydra lets users describe the outcome they want instead of writing scraping logic.

How it works

1. Describe	Enter an authorized website URL and a plain-English command.
2. Explore	TinyFish opens a browser and navigates the live website.
3. Structure	Groq converts the browser result into clean JSON data.
4. Publish	Hydra saves the workflow and creates a reusable GET or POST endpoint.

One-line pitch

Hydra turns a natural-language website instruction into a reusable JSON API - without requiring the user to write a scraper.

Core capabilities

Natural-language commands	Users describe the browser task instead of writing selectors or automation code.	**Live browser navigation**	TinyFish handles dynamic pages, navigation, and JavaScript-rendered content.
Structured responses	Groq normalizes browser results into predictable JSON for downstream applications.	**Reusable APIs**	Every successful workflow receives a GET or POST endpoint that can be called again.
Workflow dashboard	Users can create, run, inspect, version, and remove workflows from one interface.	**Read-only safety**	Hydra requires authorization and is designed for public or permitted data access.

Technology stack

Frontend	React, TypeScript, Vite
Backend	Python, FastAPI, REST endpoints
Database	SQLite workflow, version, run, and event storage
Automation	TinyFish Agent and browser/CDP integration
LLM	Groq for result interpretation and JSON normalization

Working demonstrations

Books to Scrape	Returned five Travel books with title, price, availability, and rating.
ICAI	Navigated the public site and returned five notification titles with PDF links.

Responsible use

- Use Hydra only for public information or websites where the user has permission.
- Respect website terms, robots.txt rules, authentication boundaries, and rate limits.
- Stop on CAPTCHA, access-denied responses, or technical restrictions instead of bypassing them.
- Use official APIs when a platform provides an approved integration path.

Next steps

- Automatic workflow repair when page layouts change
- Scheduled runs, alerts, and monitoring
- Authentication, API tokens, and team workspaces